

# KB0003 — Active Directory Account Lockout Investigation

**Category:** Identity & Access Management

**Last Updated:** 2026-06-01

**Owner:** Infrastructure — Windows Team

---

## Overview

This article covers diagnosing and resolving mass Active Directory lockout events, scheduled task credential failures, and service account lockouts. Distinguishes between individual user issues and systemic AD problems.

---

## Symptoms

- Multiple users locked out simultaneously
  - Service accounts locking repeatedly within minutes of unlock
  - Event ID 4740 flooding in Windows Security logs
  - Applications failing with "Logon failure: account currently locked out" (0xC0000234)
- 

## Diagnostic Commands

### Find Lockout Source

```
# Find which DC processed the logout
Get-WinEvent -ComputerName <DC-name> -FilterHashtable @{
    LogName='Security'; Id=4740
} | Select TimeCreated, Message | FL

# Check bad password attempts
Search-ADAccount -LockedOut | Select Name, DistinguishedName,
LastLogonDate

# Find the caller computer (lockout source)
Get-ADUser <username> -Properties BadPwdCount, LockedOut,
BadPasswordTime,
    LastBadPasswordAttempt | FL
```

## Check for Credential Manager Issues

```
# On user's workstation - check cached credentials
cmdkey /list

# Remove stale cached credentials
cmdkey /delete:<target>
```

## Service Account Lockout Loop

```
# Find scheduled tasks using old credentials
Get-ScheduledTask | Where-Object { $_.Principal.UserId -like "*svc_" }
|
    Select TaskName, TaskPath, @{N='User';E={$_.Principal.UserId}}

# Find IIS app pools
Get-WebConfiguration system.applicationHost/applicationPools/add |
    Where-Object { $_.processModel.userName -like "*<username>*" }
```

---

## Resolution Steps

# Individual Lockout

1. Run `Unlock-ADAccount -Identity <username>`
2. Identify source workstation from Event 4740
3. Check Credential Manager on source workstation for stale passwords
4. Update any cached credentials

# Service Account Repeated Lockout

1. Identify all services/tasks using the account (see commands above)
2. Reset password via AD → immediately update ALL dependent services
3. Change in this order: AD → Scheduled Tasks → IIS App Pools → Windows Services
4. Consider converting to Group Managed Service Account (gMSA)

# Mass Lockout Event

1. Check for password spray attack patterns in SIEM (Sentinel/Splunk)
2. Review Azure AD Sign-in logs for suspicious IPs
3. Enable conditional access policy if not already active
4. Notify Security Ops if external attack suspected

# Escalation Matrix

| Scenario                     | Escalate To             | SLA    |
|------------------------------|-------------------------|--------|
| Single user lockout          | L1 can resolve          | 30 min |
| Service account lockout loop | L2 Windows team         | 1 hour |
| Mass lockout (>10 users)     | Security Ops + AD Admin | 15 min |
| Suspected password spray     | SOC immediately         | 5 min  |

# Related Articles

- KB0001 — Password Reset & Account Unlock Procedures

- KB0002 — MFA Enrollment and Troubleshooting